

# Comparative Assessment: MENTALCLOUDS vs. SAMSUM

Deployment context: South Asia and MENA NLP/AI Practitioner Cohort — MentalCLOUDS Assessment

## Deployment Context

**Use case:** Researchers and developers building or evaluating AI-driven counseling support tools assess the capability of LLMs to produce faithful, coherent, and clinically appropriate summaries from counseling dialogues. The benchmark provides a standardized evaluation framework to compare multiple LLMs across summarization quality metrics relevant to the mental health domain.

**Target population:** NLP researchers and AI practitioners in South Asia and the Middle East and North Africa who are working on applied clinical NLP, digital mental health platforms, and LLM evaluation for low-resource or underrepresented languages and communities.

## Overall Risk Ratings

| Benchmark    | Risk | Avg. Score |
|--------------|------|------------|
| MENTALCLOUDS | HIGH | 2.2 / 5    |
| SAMSUM       | HIGH | 1.2 / 5    |

## Dimension Scores Comparison

| Dimension            | MENTAL CLOUDS | Rating           | SAMSUM | Rating           | $\Delta$ |
|----------------------|---------------|------------------|--------|------------------|----------|
| Input Ontology (IO)  | 2/5           | Significant gaps | 1/5    | Serious concern  | +1       |
| Input Content (IC)   | 2/5           | Significant gaps | 1/5    | Serious concern  | +1       |
| Input Form (IF)      | 4/5           | Minor gaps       | 2/5    | Significant gaps | +2       |
| Output Ontology (OO) | 2/5           | Significant gaps | 1/5    | Serious concern  | +1       |
| Output Content (OC)  | 2/5           | Significant gaps | 1/5    | Serious concern  | +1       |
| Output Form (OF)     | 1/5           | Serious concern  | 1/5    | Serious concern  | 0        |
| Average              | 2.2           |                  | 1.2    |                  | +1       |

$\Delta$  = MENTALCLOUDS score – SAMSUM score. Positive = regional benchmark scores higher.

## Overall Summaries

### MENTALCLOUDS (risk: HIGH)

MentalCLOUDS provides a structurally coherent English-only, CBT-aligned counseling-summarization benchmark whose authors are explicit about its American demographic origin [Q70] and three-aspect scope [Q69]. For the SA-MENA practitioner cohort, validity is highly use-case-dependent: South Asian academic researchers benchmarking English LLMs in CBT-aligned Indian institutional settings can derive partial signal, but four of five HIGH-priority dimensions (IO, OO, OC, OF) show substantive gaps. The output-form dimension is the most severe — ROUGE and BERTScore against English references provide zero valid signal for Arabic, Hindi, Urdu, Bengali, Farsi, or Dari output [Q22, Q23, WEB-8, WEB-10, WEB-12]. The input ontology and output ontology omit constructs (religious framing, shame/honor-protective omission, family-system dynamics, fatalistic attribution) that are clinically salient across the target region [WEB-4, WEB-5, WEB-6, WEB-7] and have not been NLP-operationalized elsewhere. Output

content reflects an exclusively India-affiliated annotator pool with confirmed absence of MENA validation [WEB-13]. The benchmark also exhibits a safety-critical gap on suicide-risk capture [Q64] that compounds with documented regional stigma [WEB-1, WEB-2, WEB-3]. The IF dimension (LOWER priority) is the strongest area of fit.

### SAMSUM (risk: HIGH)

SAMSum is a well-constructed informal English messenger-dialogue summarization benchmark that, by design and verifiable empirical sampling, contains no clinical content, no SA-MENA cultural content, no non-English language data, no clinical or regional output ontology, and uses metrics the authors themselves document as unreliable for dialogues — let alone for cross-lingual clinical evaluation. Five of six dimensions (IO, IC, OO, OC, OF) score 1; IF scores 2 because plain-text speaker-attributed structure provides limited transferable infrastructure. The benchmark is being applied as a proxy for clinical counseling dialogue summarization quality assessment in SA-MENA contexts, which is a substantial domain, cultural, linguistic, and methodological mismatch on every priority-HIGH dimension identified in the elicitation. The empirical dataset analysis directly confirms claims made in the regional YAML and adds concrete evidence of anti-clinical annotator norms (DATASET-D6, DATASET-D10, DATASET-D33) that go beyond mere absence into active misalignment.

## Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

### Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

| MENTALCLOUDS — 2/5 (Significant gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SAMSUM — 1/5 (Serious concern) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <p>The benchmark's three-component CBT-aligned taxonomy (Symptom and History, Patient Discovery, Reflecting) [Q15, Q26, Q31, Q40] is explicitly limited in scope, with the authors themselves acknowledging that 'this work explored only three aspects (counseling component) of the conversation' [Q69] and that the sessions represent 'a certain demographic region (American)' that 'may not apply to therapy counseling for other demographics' [Q70]. For the HIGH-priority IO dimension in the SA-MENA target context, multiple clinically salient categories are absent: religious framing of distress (sabr, tawakkul, karma, nazar), shame and honor-based help-seeking inhibition (izzat, ird), family-system and intergenerational dynamics, and fatalistic attribution of suffering — all documented in the regional clinical literature [WEB-4, WEB-5, WEB-6, WEB-7]. The benchmark also documents internal taxonomic weaknesses: components 'frequently overlap, posing clinical and legal problems' [Q62], and models could not identify psychotherapy types or therapy techniques [Q63], or capture suicide risk and negative histories [Q64] — a safety-critical gap that compounds with documented stigma barriers in South Asia and MENA [WEB-1, WEB-2, WEB-3]. Established Islamic psychology frameworks such as TIIP (nafs, qalb, aql, ruh) exist in the clinical literature but have not been operationalized into NLP annotation schemas [WEB-4, WEB-5].</p> | <p>SAMSum's input ontology is explicitly scoped to informal messenger chitchat — chit-chats, gossip, meeting arrangements, political discussion, and university assignment consultation [Q20] — positioned against news/scientific summarization [Q7], not clinical counseling. The benchmark contains no task category for CBT-structured counseling components, symptom elicitation, risk assessment, or culturally specific counseling acts (religious reframing, family-mediated resolution, izzat/namus-mediated indirect disclosure). Empirical sampling confirms zero clinical instances across 76 examples; the closest analogues are peer emotional support (DATASET-D4) and interpersonal relationship advice (DATASET-D5), neither of which involve therapeutic technique. Given the HIGH priority weight on IO for this deployment and the user's confirmation that both generic CBT structure and region-specific counseling components matter, the ontology fails on both axes.</p> |
| MENTALCLOUDS evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SAMSUM evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>[Q15] 'Within each conversation, three pivotal counseling components (aspects) emerge – symptom and history exploration, patient discovery, and reflective utterances.' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>[Q7] 'Major research efforts have focused so far on summarization of single-speaker documents like news (e.g., Nallapati et al. (2016)) or scientific publications (e.g., Nikolov et al. (2018)).' (p.1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <a href="#">[Q62]</a> 'However, the models did not do as well with the structure separation of the information. The sections of "symptoms and history", "patient discovery", and "reflection" frequently overlap, posing clinical and legal problems.' (p.15)        | <a href="#">[Q20]</a> 'It includes chit-chats, gossiping about friends, arranging meetings, discussing politics, consulting university assignments with colleagues, etc.' (p.2) |
| <a href="#">[Q63]</a> 'The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes.' (p.15)                                                                | <b>DATASET-D4:</b> Closest near-clinical analogue (peer emotional support) — still not professional counseling                                                                  |
| <a href="#">[Q69]</a> 'Finally, this work explored only three aspects (counseling component) of the conversation.' (p.16)                                                                                                                                            |                                                                                                                                                                                 |
| <a href="#">[Q70]</a> 'Additionally, the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics.' (p.16)                                                           |                                                                                                                                                                                 |
| <a href="#">[WEB-4]</a> TIIP (Traditional Islamically Integrated Psychotherapy) grounds intervention in four aspects of the Islamic conception of the soul (nafs, qalb, aql, ruh) — a structured Islamic psychology counseling framework relevant to MENA deployment |                                                                                                                                                                                 |
| <a href="#">[WEB-5]</a> Keshavarzi & Haque outline the four soul aspects as 'level' targets for clinical intervention                                                                                                                                                |                                                                                                                                                                                 |
| <a href="#">[WEB-6]</a> Cultural/spiritual treatment practices (ruqyah, dhikr, salah) documented as adjuncts in Muslim mental health literature                                                                                                                      |                                                                                                                                                                                 |
| <a href="#">[WEB-7]</a> Narrative review notes CBT's directive stance is documented as compatible with Arab and South Asian Muslim clinical expectations in some contexts                                                                                            |                                                                                                                                                                                 |
| <a href="#">[WEB-1]</a> BMC Psychiatry 2020 systematic review (n=6,767) — stigma toward mental illness in India higher than Western countries                                                                                                                        |                                                                                                                                                                                 |
| <a href="#">[WEB-2]</a> Pakistan NPMS 2022 — ~1 in 3 adults meet psychiatric criteria; help-seeking low partly due to stigma                                                                                                                                         |                                                                                                                                                                                 |
| <a href="#">[WEB-3]</a> Bangladesh 2023 — mental health treatment gap ~94%; stigma a primary structural barrier                                                                                                                                                      |                                                                                                                                                                                 |

## Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

| <b>MENTALCLOUDS — 2/5 (Significant gaps) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>SAMSUM — 1/5 (Serious concern) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>The 11.5K-utterance dataset drawn from 191 YouTube counseling sessions [Q12] is described by the authors as 'embracing a heterogeneous demographic spectrum' [Q13] but they explicitly clarify the sessions represent 'a certain demographic region (American)' [Q70]. For the SA-MENA cohort, this creates two overlapping content-validity issues. First, the input language is English only — there is zero Arabic, Hindi, Urdu, Farsi, Bengali, or Dari content [Q14], and no comparable Arabic counseling session summarization benchmark exists [WEB-14, WEB-15]; the closest Arabic resources (MentalQA, AraHealthQA) cover Q&amp;A classification rather than session dialogue summarization [WEB-12, WEB-14]. No Hindi/Urdu, Farsi, or Dari clinical dialogue summarization datasets have been identified. Second, even at the discourse level, American therapeutic communication norms are encoded in the dialogues, while MENA contexts specifically have a diglossic gap (counseling occurs in spoken dialect; clinical notes in MSA) that has no analog in the dataset. Per the elicitation, the mock-dialogue format moderates the cultural communication-norm gap somewhat, making IC a MODERATE rather than HIGH-priority concern, but the linguistic gap remains structural for non-English deployment.</p> | <p>Concrete dialogue instances were synthetically authored by linguists fluent in English at Samsung R&amp;D Institute Poland [Q17, Q19], reflecting their daily messenger conversation patterns. Empirical sampling confirms uniform European/North American cultural content: UK geography (DATASET-D29), US political content (DATASET-D11), US country music (DATASET-D35), Western consumer brands (DATASET-D28), Western dating norms (DATASET-D25), and a single 'Moroccan' reference framed orientalizingly by a European speaker (DATASET-D8). No Arabic, Urdu, Hindi code-switching, izzat/namus framing, religious distress framing (tawakkul, qadar), or stigma-mediated indirect disclosure is present. Combined with the user's HIGH priority weight on IC and confirmation that linguistic mismatch is the primary concern for MENA tooling, this constitutes a severe content validity failure.</p> |
| <b>MENTALCLOUDS evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>SAMSUM evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>[Q12] 'Comprising 11.5K utterances extracted from 191 counseling sessions involving therapists and patients, this dataset draws from publicly accessible platforms such as YouTube.' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>[Q17] 'Our dialogue summarization dataset contains natural messenger-like conversations created and written down by linguists fluent in English.' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>[Q13] 'Embracing a heterogeneous demographic spectrum with distinctive mental health concerns and diverse therapists, the dataset facilitates the formulation of a comprehensive and inclusive approach for researchers.' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>[Q19] 'We asked linguists to create conversations similar to those they write on a daily basis, reflecting the proportion of topics of their real-life messenger conversations.' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>[Q14] 'Utilizing pre-processed transcriptions derived from counseling videos, the constituent dialogues within the dataset exhibit a dyadic structure, exclusively featuring patients and therapists as interlocutors.' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>DATASET-D8: 'Moroccan casserole' described from external European perspective — only MENA reference is orientalizing</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>[Q70] 'Additionally, the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics.' (p.16)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>DATASET-D11: US political parody content requires Western cultural knowledge</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>[WEB-12] AraHealthQA 2025 — Arabic mental health Q&amp;A shared task using MentalQA (500 QA pairs), three subtasks; not session summarization</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>DATASET-D29: UK coastal geography distractor</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>[WEB-14] MentalQA — 500 Arabic mental health Q&amp;A pairs, classification taxonomy; not dialogue summarization</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>DATASET-D25: Online dating normalized without family/stigma awareness — Western individualist assumption</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

[WEB-15] MDPI scoping review confirms Arabic mental health NLP dominated by social media detection; limited labeled clinical dialogue data and dialectal variation challenges

## Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

| MENTALCLOUDS — 4/5 (Minor gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SAMSUM — 2/5 (Significant gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <p>The benchmark input is text-only transcriptions of counseling video dialogues [Q14], described as 'incoherent and grammatically unfluent' reflecting natural conversational register [Q61]. Per the dimension priority weights, IF is LOWER priority because both benchmark and the core English-system deployment use case are text-only — no modality mismatch exists for English-interface deployment. The infrastructure context confirms the benchmark's open-source &lt;7B-parameter focus [Q67, Q68] is practically applicable across SA-MENA compute environments. The remaining IF concern is script and right-to-left rendering: the benchmark provides no Arabic, Urdu, or Persian-script input form, but this is more accurately a content/language gap (covered in IC) than a signal-encoding mismatch. For the LOWER-priority IF dimension, this score reflects substantive alignment for English-system deployment with minor gaps for multilingual extension.</p> | <p>The plain-text turn-by-turn speaker-attributed format [Q29, Q33] is structurally compatible with clinical NLP pipelines and is empirically clean and consistent (DATASET-D17, DATASET-D19). Speaker attribution and utterance separators have been shown to benefit models [Q65, Q66]. However, all formatting decisions, tokenization, and length calibrations [Q42] assume English plain text exclusively. There is no provision for Arabic RTL rendering, Urdu Nastaliq, Devanagari, morphologically rich tokenization, or cross-script code-switching common in South Asian clinical contexts. Empirical sampling confirms 100% English Latin-script content with no non-Latin or RTL examples (DATASET-D17, DATASET-D19). The MODERATE priority weight reflects partial transferability of structural conventions with substantive multilingual mismatches.</p> |
| MENTALCLOUDS evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | SAMSUM evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>[Q14] 'Utilizing pre-processed transcriptions derived from counseling videos, the constituent dialogues within the dataset exhibit a dyadic structure, exclusively featuring patients and therapists as interlocutors.' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>[Q29] 'Beforehand, we specified a format for written dialogues with summaries: a colon should separate an author of utterance from its content, each utterance is expected to be in a separate line.' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>[Q61] 'Hence, most of these natural conversations are incoherent and grammatically unfluent.' (p.15)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>[Q33] 'Each utterance contains the name of the speaker.' (p.3)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>[Q67] 'Second, for faster and easier reproduction, we did not assess models larger than 7 billion parameters; however, such models can be part of future examinations.' (p.16)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>[Q42] 'In all our experiments, news and dialogues are truncated to 400 tokens, and summaries – to 100 tokens.' (p.4)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>[Q68] 'Third, for the initial study and to promote research in this field, only open-source models were assessed in this work.' (p.16)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>[Q65] 'On the other hand, dialogues are divided into utterances, and for each utterance its author is assigned.' (p.7)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>[Q66] 'We demonstrate in experiments that the models benefit from the introduction of separators, which mark utterances for each person.' (p.7)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>DATASET-D17: Consistent format with semi-formal register and repeated-speaker turns</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>DATASET-D19: Clean three-party speaker-colon-utterance format</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

| <b>MENTALCLOUDS — 2/5 (Significant gaps) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>SAMSUM — 1/5 (Serious concern) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <p>The output label space comprises (a) free-text aspect-based summaries against the three counseling components [Q19], (b) the six-parameter Sekhon clinical acceptability framework — affective attitude, burden, ethicality, coherence, opportunity costs, perceived effectiveness [Q44, Q48], and (c) a three-level hallucination scale [Q46, Q53, Q54]. For the HIGH-priority OO dimension, the label categories encode CBT-aligned Indian/US professional consensus about what counts as a clinically appropriate, coherent, or effective summary. Parameters such as 'perceived effectiveness,' 'ethicality,' and 'opportunity costs' are normative judgments that depend on the professional clinical frame applied [Q51]; MENA practitioners integrating Islamic psychology frameworks (TIIP, ruqyah/dhikr/salah adjuncts) [WEB-4, WEB-5, WEB-6] may calibrate these parameters differently. The AraHealthQA 2025 work explicitly noted that surface-level metrics 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA [WEB-12], supporting the structural concern that the OO is calibrated to a specific professional frame. The authors themselves acknowledge models score poorly on 'overall efficacy and the opportunity cost' [Q51] and that important negative histories (e.g., suicide risk) are not recorded [Q64] — a safety-critical taxonomic absence. There is no output category for religious framing, family-system salience, shame/honor-protective omissions, or fatalistic attribution.</p> | <p>Each dialogue receives exactly one reference summary [Q24] under a four-point style guide (short, information-extractive, interlocutor-naming, third-person) [Q23]. There are no clinical sub-categories — no symptom identification, risk indicators, therapeutic intervention markers, or empathy/alliance categories — and no representation of region-specific counseling outputs (religious reframing, family-mediated resolution plans, stigma-sensitive omissions). Empirical sampling reveals concrete failure modes: suicide discussed as social reassurance trivia (DATASET-D10), abuse-victim self-derogation reproduced as factual summary content (DATASET-D6), peer distress reduced to social transaction (DATASET-D33). The single-reference design forecloses representation of regional practitioner norm divergence. Given HIGH priority weight on OO and user confirmation that region-specific salience matters, this is a severe structural validity violation.</p> |
| <b>MENTALCLOUDS evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SAMSUM evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>[Q19] 'This section reports the aspect-based (psychotherapy element-based) summarization results based on the automatic evaluation scores.' (p.9)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>[Q23] 'we asked language experts to annotate them with summaries, assuming that they should (1) be rather short, (2) extract important pieces of information, (3) include names of interlocutors, (4) be written in the third person.' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>[Q44] 'The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness.' (p.12)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>[Q24] 'Each dialogue contains only one reference summary.' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>[Q46] 'Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed).' (p.12)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>DATASET-D6: Stigmatizing self-description 'Pearl is fat and ugly' reproduced as factual summary — no clinical reframing category</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>[Q48] 'The clinical acceptability framework [39] involves six parameters – affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness.' (p.13)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>DATASET-D10: Suicide statistics summarized as social reassurance — no risk/safety category</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>[Q51] 'As all three models have poor scores on the more sensitive aspects i.e. the overall efficacy and the opportunity cost, this indicates that these models share the same weakness and are not suitable for clinical use as they stand now.' (p.14)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>DATASET-D33: Peer emotional support reduced to social transaction — no distress/intervention category</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| [Q53] 'Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations.' (p.14)      |  |
| [Q54] 'These categories essentially determine how well the response is consistent with the context and whether it is also incorrect, nonsensical, or contains global information beyond the scope of the conversation.' (p.14)       |  |
| [Q64] 'Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified.' (p.15) |  |
| [WEB-4] TIIP framework grounds intervention in Islamic conception of the soul (nafs, qalb, aql, ruh) — relevant clinical taxonomy not encoded in MentalCLOUDS                                                                        |  |
| [WEB-5] Keshavarzi & Haque four soul aspects as 'level' targets for Islamic-integrated clinical intervention                                                                                                                         |  |
| [WEB-6] Ruqyah, dhikr, salah documented as cultural/spiritual treatment adjuncts in Muslim mental health literature                                                                                                                  |  |
| [WEB-12] AraHealthQA 2025 found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA                                                                                                   |  |



## Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

| <b>MENTALCLOUDS — 2/5 (Significant gaps) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>SAMSUM — 1/5 (Serious concern) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <p>Reference summaries were produced by augmenting MEMO with annotated dialogue summaries [Q17], with limited documentation of the initial labelers. The qualitative-evaluation expert pool is fully India-affiliated: five healthcare professionals (two clinical psychologists, three psychiatrists/medical practitioners) [Q41, Q42], aged 40–55, four male and one female [Q43], from IIT Delhi, AIIMS Rishikesh, and AIIMS New Delhi [Q9]. There is no documented MENA representation, and web searches confirm that 'no MENA-based mental health professional validation of MentalCLOUDS annotations has been conducted or announced' [WEB-13]. For the HIGH-priority OC dimension, this means salience judgments embedded in reference summaries reflect CBT-aligned Indian/US professional consensus and may not correlate with MENA professional judgments — a convergent-validity concern. The paper documents inter-rater variance ('raters were more aligned in rating MentalBART... with lesser variance' [Q50]; 'fluctuations in how hallucinations are perceived among different models' [Q57]) but does not analyze this variance by professional background or regional training. Aggregation by simple averaging across five raters [Q47] further risks erasing minority perspectives, though the small pool limits the severity of that concern. Notably, the affiliated practitioner organization YourDOST [Q9] strengthens India-specific OC validity for South Asian academic users somewhat but does not extend to MENA contexts.</p> | <p>Reference summaries were authored by the same linguists who created the dialogues [Q23], creating data-label circularity, with a 50-conversation double-annotation validation showing 94% agreement [Q26] but limited generalization. Annotator demographics are not reported; acknowledged contributors have Polish-consistent names [Q75]. The user elicitation explicitly confirms annotators were 'general mental health experts from mixed backgrounds (including India and the USA)' but 'were not specifically briefed on regional context when validating summaries' and that 'their salience judgments likely reflect generic CBT protocols rather than consciously incorporating regional norms.' Empirical evidence reveals concrete content failures: a stigmatizing self-description reproduced as factual content (DATASET-D6), a speaker attribution error in a simple two-turn exchange (DATASET-D21), peer distress treated as social transaction (DATASET-D33). HIGH priority weight on OC and confirmed norm-divergence risk produce a severe convergent validity failure.</p> |
| <b>MENTALCLOUDS evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SAMSUM evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>[Q9] 'Department of Electrical Engineering, Indian Institute of Technology Delhi, India; ... All India Institute of Medical Sciences, Rishikesh, India; ... All India Institute of Medical Sciences, New Delhi, India; Yardi School of Artificial Intelligence, Indian Institute of Technology Delhi, India' (p.1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>[Q23] 'we asked language experts to annotate them with summaries' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>[Q17] 'Expanding upon the MEMO dataset, we augment it with annotated dialogue summaries corresponding to the three identified' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>[Q26] '94% of examined dialogues were classified by both annotators as good' (p.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>[Q41] 'In order to conduct a comprehensive expert assessment, five healthcare professionals were employed to assess the clinical appropriateness of the summaries produced by the LLMs based on the evaluation framework of Sekhon et al.' (p.12)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>[Q75] 'sincere thanks to Tunia Błachno, Oliwia Ebebenge, Monika Jędras and Małgorzata Krawentek' (p.9)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>[Q42] 'Among them were two clinical psychologists, with the remaining three comprising psychiatrists and medical practitioners.' (p.12)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>DATASET-D6: Stigmatizing self-description reproduced as factual summary content — concrete annotator norm divergence</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>[Q43] 'Of the group, four were male, and one was female, with ages ranging from 40 to 55 years and possessing over a decade of therapeutic experience.' (p.12)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>DATASET-D21: Speaker attribution error (Miranda vs. Danny) — undermines 94% quality ceiling</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



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| <a href="#">[Q47]</a> 'Table 6 reports the clinical experts' scores averaged over the ratings given by five experts.' (p.13)                                                                                                   | <a href="#">DATASET-D33</a> : Distress disclosure summarized as social transaction — anti-clinical norm embedded in label |
| <a href="#">[Q50]</a> 'Overall, all the raters were more aligned in rating the MentalBART model with lesser variance as compared to the other two LLMs for all the metrics.' (p.13)                                            |                                                                                                                           |
| <a href="#">[Q57]</a> 'The data shows fluctuations in how hallucinations are perceived among different models and stresses the importance of reviewing evaluations from numerous appraisers for a complete assessment.' (p.14) |                                                                                                                           |
| <a href="#">[WEB-13]</a> JMIR Mental Health 2024 — confirms no MENA-based mental health professional validation of MentalCLOUDS annotations conducted or announced; expert pool exclusively India-affiliated                   |                                                                                                                           |

## Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

| <b>MENTALCLOUDS — 1/5 (Serious concern) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>SAMSUM — 1/5 (Serious concern) [high confidence]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p>Automatic evaluation uses ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore <a href="#">[Q20, Q21]</a>, all computed against English reference summaries via 'overlap of n-grams... between the generated summary and reference summaries' <a href="#">[Q22, Q23]</a> and 'longest co-occurring n-gram' <a href="#">[Q24]</a>. For the HIGH-priority OF dimension, this design provides zero valid evaluation signal for systems generating summaries in Arabic, Hindi, Urdu, Bengali, Farsi, or Dari — covering the majority of MENA-focused practitioners and many South Asian sub-populations. The 2026 multilingual mental health evaluation study confirms that for Arabic, BLEU scores ~0.54 vs LaBSE ~0.78 demonstrate n-gram metrics are 'especially poorly suited for Arabic clinical NLP evaluation' <a href="#">[WEB-8]</a>. The XLEval benchmark documents statistically significant English-vs-Hindi BERTScore performance gaps confirming validity concerns for non-English clinical NLP <a href="#">[WEB-10]</a>. AraHealthQA 2025 explicitly found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA <a href="#">[WEB-12]</a>. Cross-lingual alternatives (LaBSE, XLM-RoBERTa BERTScore, Arabic-specific BERTScore via AraBERT/CAMeLBERT/MARBERT, Hindi/Urdu via MuRIL/IndicBERT) are technically viable <a href="#">[WEB-8, WEB-9, WEB-11]</a> but have not been applied to MentalCLOUDS or to clinical summarization in target languages. The qualitative six-parameter expert evaluation <a href="#">[Q45]</a> is partially region-transferable as scaffolding but inherits the calibration concerns documented in OO. Per A4, the benchmark 'is applicable only to English-language systems; multilingual evaluation requires an entirely separate framework.'</p> | <p>Primary evaluation uses ROUGE F1 (R-1, R-2, R-L) <a href="#">[Q47]</a> supplemented by a –1/0/1 human rating scale <a href="#">[Q48]</a>. The benchmark itself documents that ROUGE is misleading for dialogue summarization — high ROUGE corresponds with low human ratings <a href="#">[Q3, Q54, Q67, Q68]</a> — and explicitly calls for new metrics <a href="#">[Q69, Q74]</a> without providing them. For SA-MENA deployment, this dialogue-summarization unreliability is compounded by complete inapplicability to non-English outputs: ROUGE/BERTScore against English references provide no valid signal for Arabic, Hindi, Urdu, Farsi, or Dari per the regional YAML and user elicitation. <a href="#">WEB-13</a> (AraHealthQA 2025) directly corroborates that BERTScore fails to capture clinical appropriateness for Arabic mental health tasks. Empirical sampling confirms 100% English content (<a href="#">DATASET-D19, DATASET-D17</a>), with no multilingual signal whatsoever. HIGH priority weight on OF and complete cross-lingual misalignment yield a severe external validity failure.</p> |
| <b>MENTALCLOUDS evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>SAMSUM evidence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| <a href="#">[Q20]</a> 'Given the generative nature of the task, we employ standard summarization evaluation metrics such as Rouge-1 (R-1), Rouge-2 (R-2), Rouge-L (R-L), and BERTScore (BS) along with their corresponding Precision (P), Recall (R) and F1 scores.' (p.9) | <a href="#">[Q3]</a> 'We show that model-generated summaries of dialogues achieve higher ROUGE scores than the model-generated summaries of news – in contrast with human evaluators' judgement.' (p.1)           |
| <a href="#">[Q21]</a> 'Since F1 accounts for Precision and Recall, we compare LLM's performance based on F1 unless stated otherwise.' (p.9)                                                                                                                                | <a href="#">[Q47]</a> 'We evaluate models with the standard ROUGE metric (Lin, 2004), reporting the F1 scores (with stemming) for ROUGE-1, ROUGE-2 and ROUGE-L' (p.4)                                             |
| <a href="#">[Q22]</a> 'ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [58] assesses the overlap of n-grams (sequences of n consecutive words) between the generated summary and reference summaries.' (p.9)                                                     | <a href="#">[Q67]</a> 'We show that the most popular summarization metric ROUGE does not reflect the quality of a summary.' (p.8)                                                                                 |
| <a href="#">[Q23]</a> 'This metric measures the number of overlapping units such as n-gram, word sequences, and word pairs between the generated summary evaluated against the gold summary typically created by humans.' (p.9)                                            | <a href="#">[Q69]</a> 'a new metric should be designed to measure the quality of abstractive dialogue summaries.' (p.8)                                                                                           |
| <a href="#">[Q24]</a> 'ROUGE-L takes into account the longest co-occurring n-gram between the candidate and the reference summaries.' (p.9)                                                                                                                                | <a href="#">[Q74]</a> 'In order to perform well, it may require designing dedicated tools, but also new, non-standard measures' (p.9)                                                                             |
| <a href="#">[Q45]</a> 'Experts evaluate each summary against these acceptability parameters, assigning continuous ratings on a scale from 0 to 2, where a higher rating signifies enhanced acceptability.' (p.12)                                                          | <a href="#">[WEB-13]</a> AraHealthQA 2025 found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA — direct corroboration that automated metrics are insufficient |
| <a href="#">[WEB-8]</a> arXiv 2602.02440 — Arabic LLM evaluation: BLEU ~0.54 vs LaBSE ~0.78 confirms n-gram metrics like ROUGE poorly suited for Arabic clinical NLP                                                                                                       | <b>DATASET-D19:</b> Entirely English content representative of 100% monolingual character                                                                                                                         |
| <a href="#">[WEB-9]</a> MDPI Healthcare 2025 — AraBERT, CAMELBERT, MARBERT applied to Arabic mental health classification; BERTScore for Arabic clinical summarization not yet evaluated                                                                                   | <b>DATASET-D17:</b> No code-switching or non-Latin script across 76 sampled examples                                                                                                                              |
| <a href="#">[WEB-10]</a> ACM Web Conference 2024 (X LingEval) — statistically significant BERTScore gap between English (0.9206) and Hindi for healthcare queries, confirming metric validity concerns for non-English clinical NLP                                        |                                                                                                                                                                                                                   |
| <a href="#">[WEB-11]</a> MDPI Applied Sciences 2025 — XLM-RoBERTa BERTScore demonstrated robust for cross-lingual summarization including low-resource languages; not yet applied to clinical mental health summarization                                                  |                                                                                                                                                                                                                   |
| <a href="#">[WEB-12]</a> AraHealthQA 2025 — BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA                                                                                                                             |                                                                                                                                                                                                                   |